HEATER v1.0.0 — Complete Data Source Overview

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Summary

HEATER pulls from **8 external APIs**, **3 web scrapes**, and **2 locally computed systems**, totaling **36 distinct data feeds**. All external calls follow a "live-first with graceful degradation" pattern: API → session cache → SQLite fallback → sensible defaults.

External API Sources

1. MLB Stats API (statsapi)

#	Endpoint	What It Fetches	Key Data Fields	Storage	Staleness	Consum
1	`sports_players`	All MLB players (9K+)	mlb_id, name, team, positions, bats, throws, birth_date	`players` table	7 days	All pages vi `load_player`
2	`team_roster` (hydrated)	Current season stats	PA, AB, H, R, HR, RBI, SB, AVG, OBP, IP, W, L, SV, K, ERA, WHIP	`season_stats` table	1 hour	My Team, T Analyzer, Pl Compare, L
3	`team_roster` (per year)	Historical stats (3 years)	Same as above, per season	`season_stats` table	30 days	Backtesting model, Mar
4	`schedule`	Today's games	game_pk, home/away teams, game_datetime, venue	In-memory	2 hours	Game-day intelligence Optimizer
5	`schedule` (hydrated)	Probable pitchers	pitcher name, mlb_id, team	In-memory DataFrame	2 hours	Start/Sit, Optimizer, Matchup Pl
6	`team_stats` (hitting)	Team batting strength	wRC+, K%, BB%, ISO, wOBA, OPS	`team_strength` table	24 hours	Game-day, Matchup Pl Streaming
7	`team_stats` (pitching)	Team pitching strength	ERA, WHIP, FIP, K/9, BB/9	`team_strength` table	24 hours	Game-day intelligence

8	`person` (hydrated)	Player enhanced status	IL status, roster_type, transactions	`player_news` table	On-demand	Injury mod Player card
9	`transactions`	Recent MLB transactions	player, type, team_from/to, date	`player_news` table	1 hour	News aggregatio Manager
10	`team_roster` (40Man)	40-man roster status	mlb_id, roster_type flag	`players` table	7 days	Player eligi
11	`/people/{id}/stats`	MiLB stats for prospects	AVG, OBP, SLG, HR, SB, K%, BB% (hit); IP, ERA, WHIP, K/9 (pitch)	`prospect_rankings` table	7 days	Leaders/Pr page

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2. Yahoo Fantasy API

#	Endpoint / Method	What It Fetches	Key Data Fields	
12	`get_league_teams` + `get_team_roster`	All 12-team rosters	team_name, player_name, position, status, injury_note, percent_owned, FAAB	`league` `league` sessio
13	`get_league_standings`	Standings + category totals	rank, W-L-T, R/HR/RBI/SB/AVG/OBP/W/L/SV/K/ERA/WHIP totals	`league` table;
14	`get_league_scoreboard_by_week`	Current weekly matchup	week, user/opp scores per category, wins/losses/ties	sessio
15	REST `/players;status=FA`	Free agent pool	player_name, positions, team, percent_owned	sessio fallba
16	`get_league_transactions`	Recent league transactions	type (add/drop/trade), player, teams, timestamp	sessio

17	`get_league_settings`	League configuration	league_name, num_teams, categories, roster_positions	sessio
18	`get_league_scoreboard` (all weeks)	Full season schedule	week to matchup pairs for all 24 weeks	`league` table
19	`get_league_draft_results`	Draft picks	pick_number, round, team, player	`league` table

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3. FanGraphs API

#	Endpoint	What It Fetches	Key Data Fields	Storage	Stal
20	`/api/projections` (7 systems)	Projections: Steamer, ZiPS, Depth Charts, ATC, TheBat, TheBatX + 3 ROS	PA, AB, H, R, HR, RBI, SB, AVG, OBP (hit); IP, W, L, SV, K, ERA, WHIP, FIP (pitch)	`projections` table (per system)	24 h
21	Embedded in projections JSON	FanGraphs ADP	player_name, adp	`adp` table	24 h
22	`/api/prospects/board/data`	Prospect rankings + scouting	fg_rank, fg_fv (20-80), fg_eta, fg_risk, scouting grades (hit/game/raw/speed/field/ctrl)	`prospect_rankings` table	7 da

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4. pybaseball (FanGraphs + Baseball Savant)

#	Function	What It Fetches	Key Data Fields	Storage	Staleness	Consumers
23	`team_batting(season)`	Team batting stats	wRC+, OPS, K%, BB%, AVG, SLG	`team_strength` table	24 hours	Game-day, Matchup adjustments
24	`team_pitching(season)`	Team pitching stats	ERA, WHIP, FIP, K/9, BB/9	`team_strength` table	24 hours	Game-day, Matchup adjustments
25	`statcast_batter()`	Statcast batted ball data	EV, barrel%, hard_hit%, xBA, xSLG, xwOBA, LA sweet spot%, pull%	`statcast_archive` table	24 hours	Trade engine signals, Player Compare

26	`statcast_pitcher()`	Statcast pitch-level data	pitch_mix %, velocities, spin_rate, swinging_strike%	`statcast_archive` table	24 hours	Trade engine signals
27	Park factor tables	Park factors	team_code, factor_hitting, factor_pitching	`park_factors` table	30 days	Optimizer, Contextual factors

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5. Other External Sources

#	Source	What It Fetches	Key Data Fields	Storage	Staleness	Consumers
28	**Open-Meteo API**	Game-day weather (30 stadiums)	temp_f, wind_speed/direction, precipitation%, humidity%	`game_day_weather` table	2 hours	Optimizer (HR adj via Nathan physics), Start/Sit
29	**ESPN Injuries API**	Real-time injury feed	player, status (IL/DTD), injury_type, body_part, return_date	`player_news` table	1 hour	Closer Monitor, Player cards, IL Manager
30	**Roster Resource** (scrape)	Depth charts (30 teams)	Batting order 1-9, rotation 1-5, bullpen roles (CL/SU/MR), committee flags	In-memory dict	7 days	Closer Monitor, Contextual factors
31	**Baseball Reference** (scrape)	Contract year players	Set of player names in upcoming FA class	In-memory set	30 days	Draft value adjustments
32	**FantasyPros** (scrape)	Expert consensus rankings	ecr_rank, team, position	Not persisted	24 hours	ECR consensus (non-functional: needs JS)
33	**NFBC** (scrape)	NFBC ADP	player_name, nfbc_adp	Not persisted	24 hours	ECR consensus (non-functional: needs JS)

Locally Computed Data Sources

#	Source	Method	Key Data Fields	Storage	Consumers

34	**Marcel Projections**	3-yr weighted regression (5/4/3), age curve, regression to mean	All counting + rate stats	In-memory dict	Fallback when FanGraphs unavailable
35	**Blended Projections**	Ridge regression stacking of all 7+ systems	All stat columns, reliability weight	`projections` table (system='blended')	Default projection source for ALL analysis
36	**ECR Consensus**	Trimmed Borda Count of sources 21, 32, 33 + ESPN/CBS/Yahoo + HEATER SGP	consensus_avg, rank_min/max, stddev, n_sources	`ecr_consensus` table	Rankings, Draft simulator

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Data Flow Architecture

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External APIs ──► Bootstrap Pipeline (21 phases) ──► SQLite Database
                                                        │
Yahoo Fantasy API ──► 3-Tier Cache ──► session_state ───┤
                                                        │
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                                              │                    │
                                        Streamlit Pages      Engine Modules
                                        (11 pages)           (Trade, Optimizer,
                                                              Draft, Matchup)
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Cache Tiers

1. **Tier 1:** `st.session_state` — fastest, per-session
2. **Tier 2:** Yahoo/MLB/FG API — live source of truth
3. **Tier 3:** SQLite database — persistent fallback

Bootstrap Pipeline Order

Phases run with staleness checks (skipped if fresh). `Force Refresh` bypasses all checks.

1. Players (MLB API) — must come first
2. Park Factors + Projections (parallel)
3. Live Stats + Historical (parallel)
4. Injury Data (derived from historical)
5. Yahoo League Data
6. Extended Roster (40-man)
7. Multi-source ADP
8. Depth Charts
9. Contract Data
10. News/Transactions
11. Deduplication
12. Prospect Rankings
13. News Intelligence

14. ECR Consensus
15. Player ID Map
16. Yahoo Transactions Sync
17. Blended Projections
18. Statcast Data
19. Game-Day Intelligence

SQLite Tables Summary

Table	Primary Source(s)	Key Purpose
`players`	MLB Stats API	Master player registry (9K+)
`projections`	FanGraphs (7 systems) + Marcel + Blended	All projection data
`season_stats`	MLB Stats API	Current + historical season stats
`adp`	FanGraphs, Yahoo, FantasyPros, NFBC	Average draft position
`park_factors`	pybaseball / FanGraphs	Venue hitting/pitching adjustments
`team_strength`	pybaseball / MLB Stats API	Team offensive/pitching quality
`statcast_archive`	pybaseball (Baseball Savant)	Batted ball + pitch metrics
`injury_history`	Derived from historical stats	Games played, IL stints
`prospect_rankings`	FanGraphs Board + MiLB stats	Prospect scouting + readiness
`ecr_consensus`	Multi-source blend	Expert consensus rankings
`player_news`	ESPN, MLB, Yahoo	Injuries, transactions, news
`game_day_weather`	Open-Meteo	Stadium weather forecasts
`league_rosters`	Yahoo Fantasy API	All 12-team rosters
`league_standings`	Yahoo Fantasy API	Category standings
`league_schedule_full`	Yahoo Fantasy API	Season matchup schedule
`league_draft_picks`	Yahoo Fantasy API	Draft results
`league_teams`	Yahoo Fantasy API	Team metadata